ISEP INSTITUTO SUPERIOR DE ENGENHARIA DO PORTO

INFME – PL4

Node.JS Routing | Luís Conceição, Diogo Martinho, Vítor Crista

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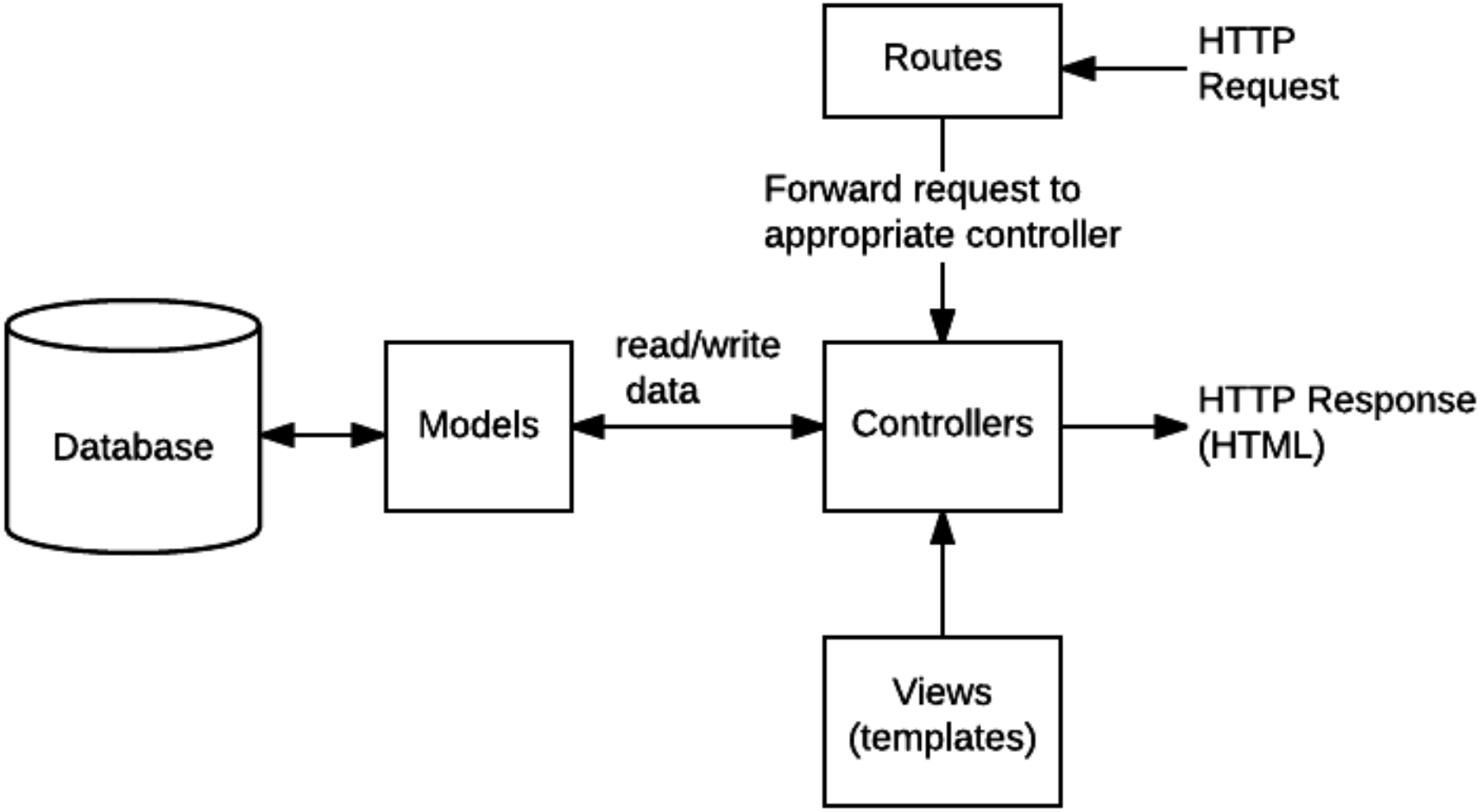
Routing



cf. https://developer.mozilla.org/en-US/docs/Learn/Server-side/Express_Nodejs/routes

Defining Routes

- The **express** module provides routing functionality;
- The express object has a **`Router()`** method that creates a router object;
- Middleware and HTTP method routes (such as `get`, `put`, `post`) can be added to router object.

```
var router = express.Router();
router.use(function(req, res, next) {
    console.log('Something is happening.');
    next();
});

router.get('/', function(req, res) {
    res.json({ message: 'hooray! welcome to our api!' });
});
```

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Defining Routes

- The `route(<path>)` method returns an instance of a single route, which can be used to handle HTTP verbs with optional middleware.

```
var router = express.Router();

router.route('/tasks')
    .get(todoList.list_all_tasks)
        .post(todoList.create_a_task);

router.route('/tasks/:taskId')
    .get(todoList.read_a_task)
    .put(todoList.update_a_task)
    .delete(todoList.delete_a_task);
```

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Tutorial 2 - Node.JS

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Tutorial 2 NodeJS

- Building a RESTful API with Node and the Express library;

- Create a Node.js project;

- Create a database (MongoDB);
- Define routes and handle HTTP requests (Express);
- Perform CRUD operations using HTTP methods;
- The app always returns JSON objects;

- Testing requests with Bruno.

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Creating a Node.js project for tutorial 2

- Create a folder named “app” in any directory and add a file called “server.js”;
- Inside the “app” folder, create another folder called “models”.
- Inside this new folder, add a file called “patients.js”;

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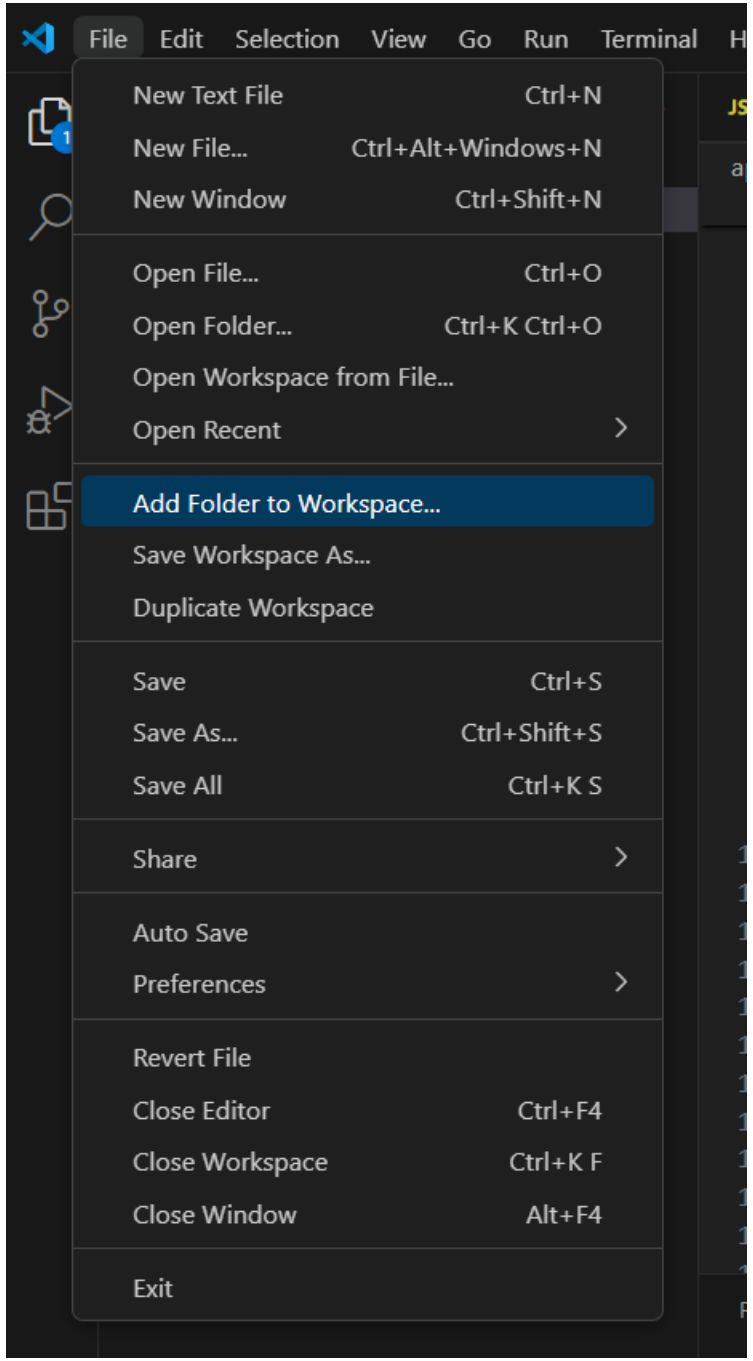
- Using Visual Studio Code, import the "app" folder into the workspace:

"File" -> "Add Folder to Workspace..."



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Creating a Node.js project

- node-project
  - node_modules
  - package-lock.json
  - package.json

- **First Step:** Create a folder called “ProjectFolder”, this will be the directory of the app
- **Second Step:** `npm init -y` to generate the `package.json`
- **Third Step:** install the necessary dependencies for this Project:
  - Express, Mongoose, Body-parser with the command:
    - `npm install express mongoose body-parser cors`

```
1   {
2     "name": "node-project",
3     "version": "1.0.0",
4     "description": "",
5     "license": "ISC",
6     "author": "",
7     "type": "commonjs",
8     "main": "server.js",
      ▷Debug
9     "scripts": {
10      "test": "echo \"Error: no test specified\" && exit 1",
11      "start": "node server.js"
12    },
13    "dependencies": {
14      "body-parser": "^2.2.2",
15      "cors": "^2.8.6",
16      "express": "^5.2.1",
17      "mongoose": "^9.3.1"
18    }
19  }
20
```

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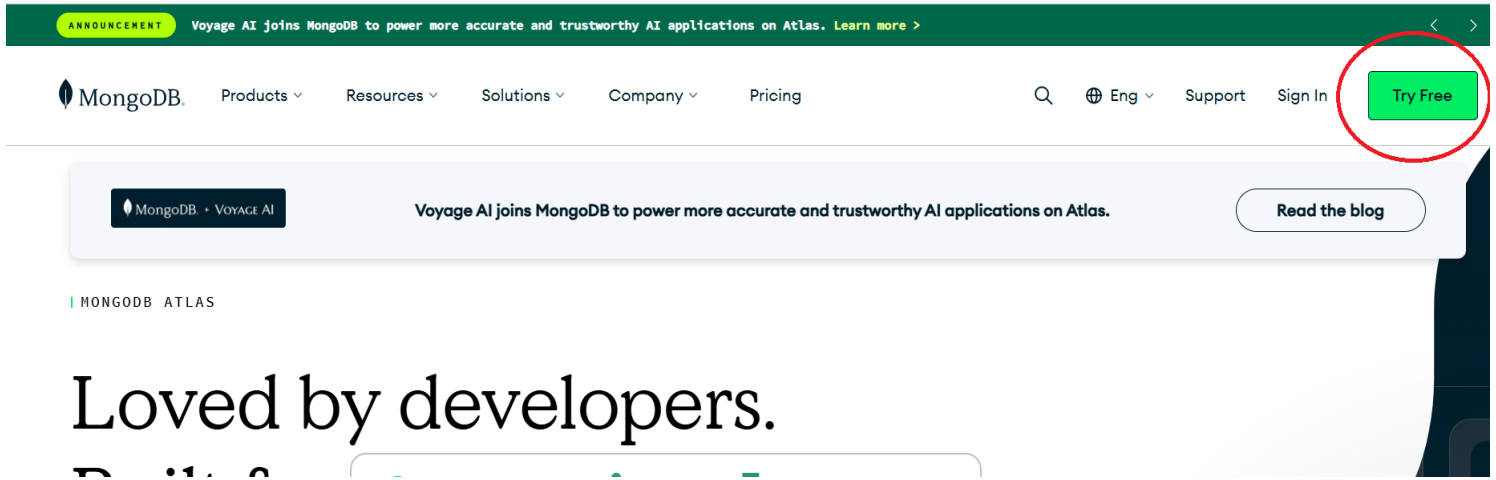
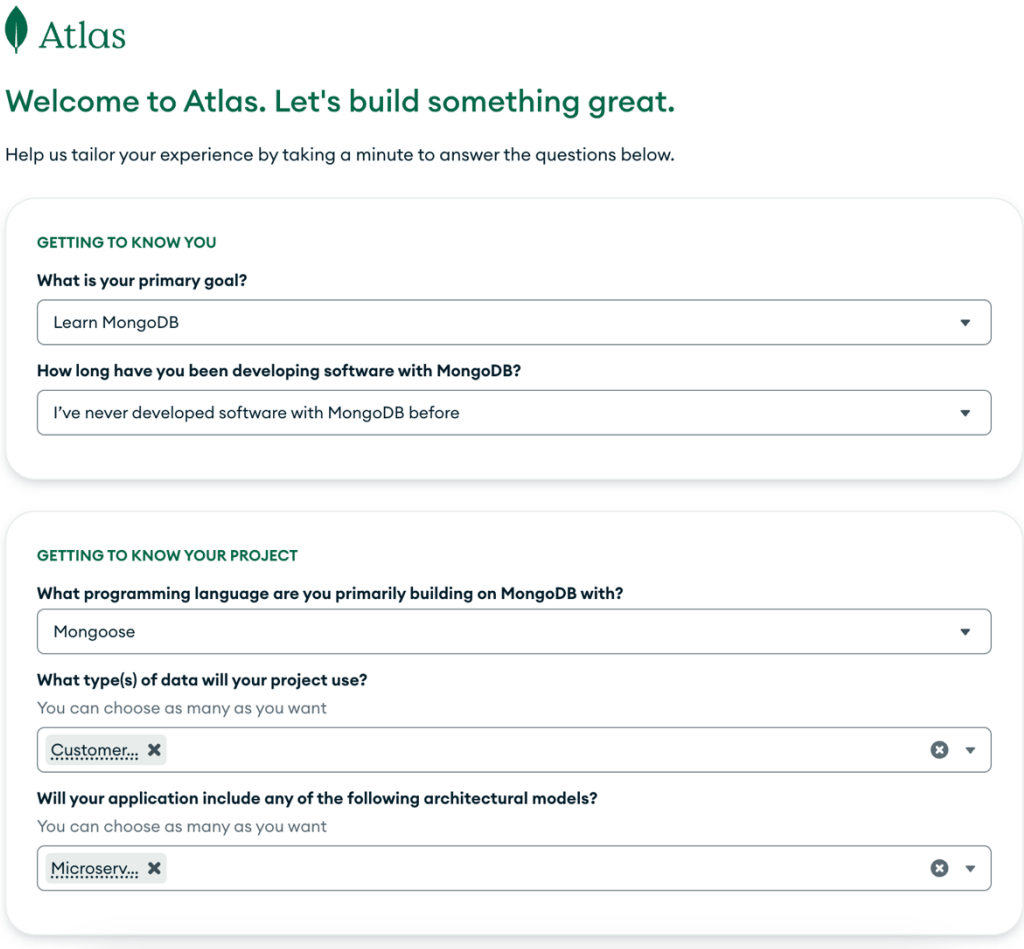
Database

- Go to the MongoDB site (https://www.mongodb.com/) and create a free account



- Use an email address (institutional or personal), validate the email, and log in.

Get Started with Mongodb



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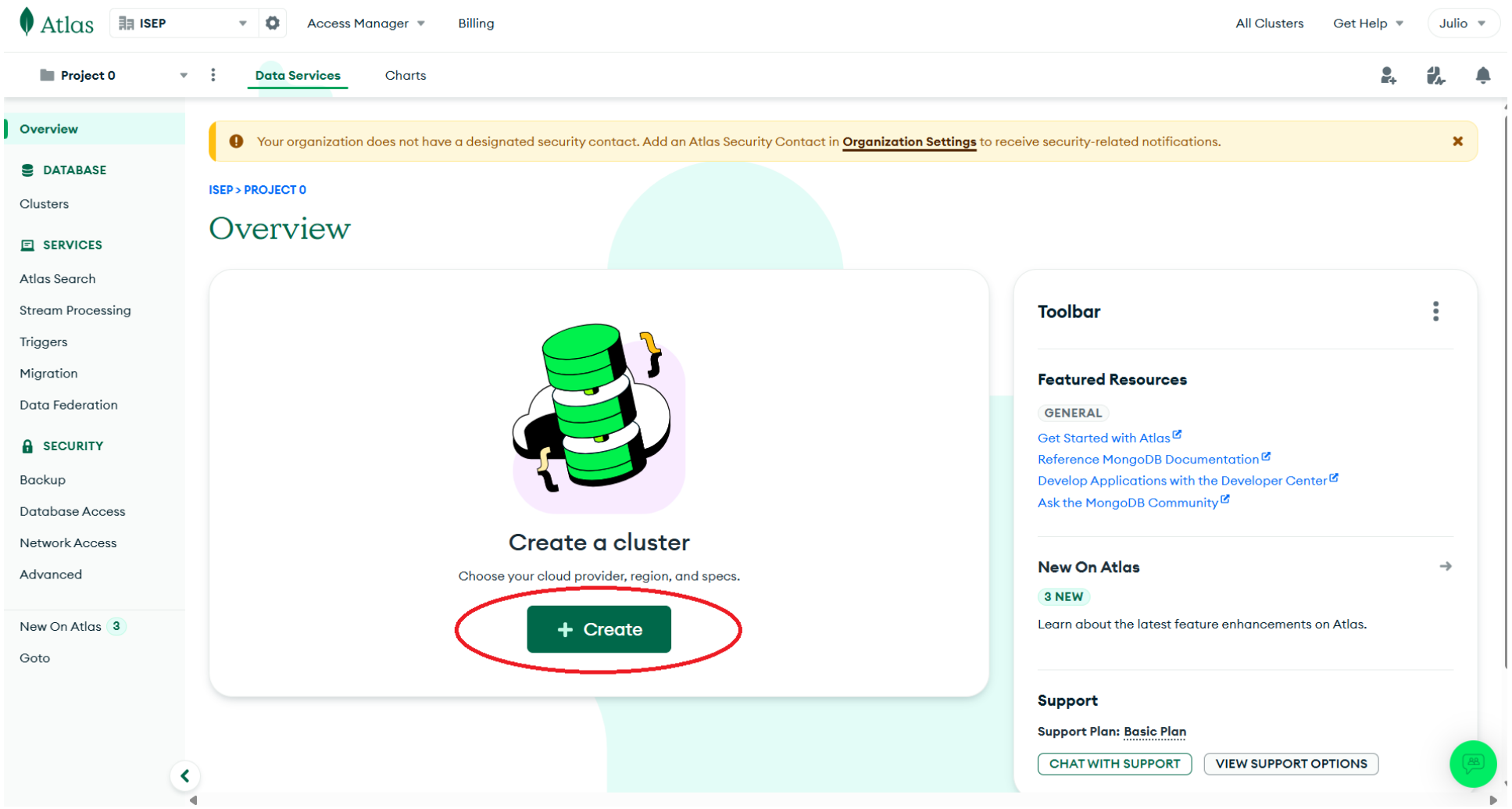
Database – Create a Cluster



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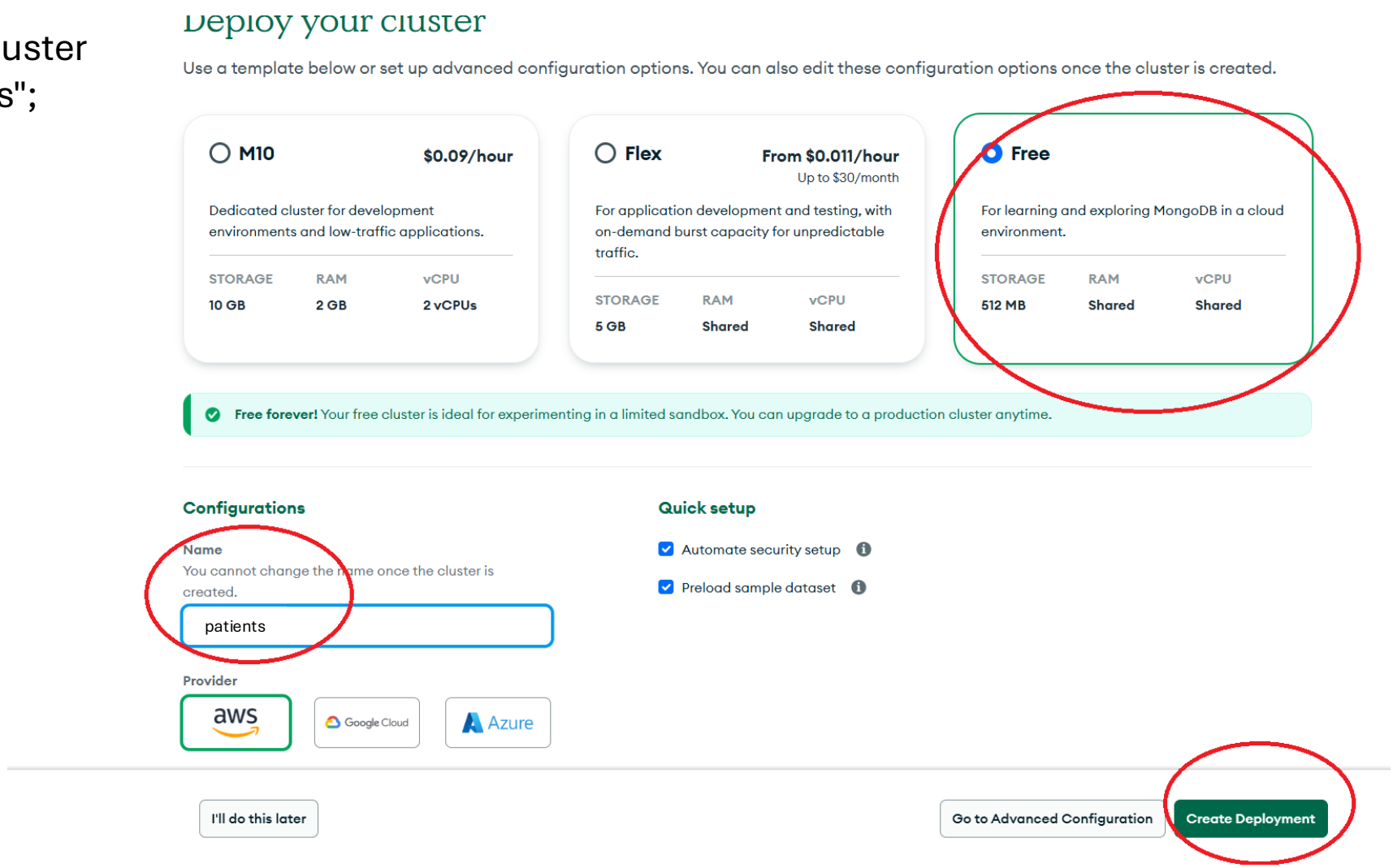
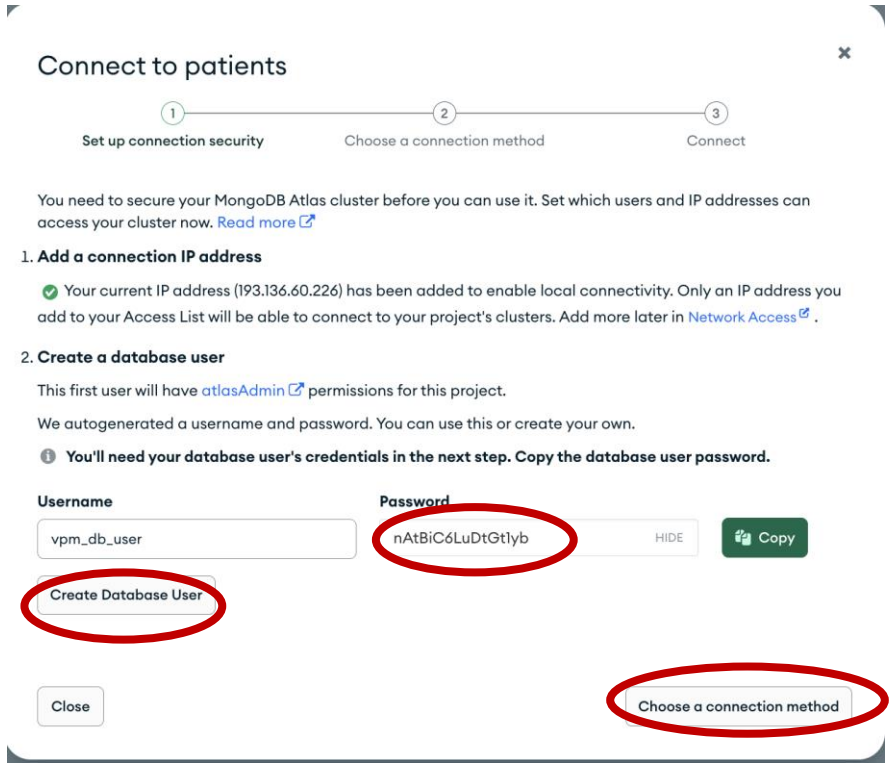
Database – Create a Cluster

Create a free cluster called ”patients";



Database – Configure the DB connection

- Copy the password and select “create database user” and next “Choose a connection method”



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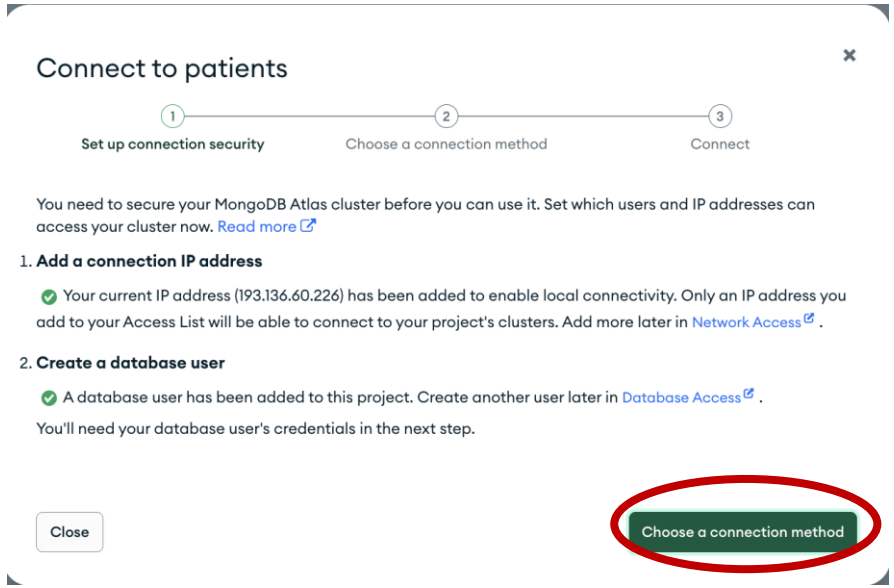

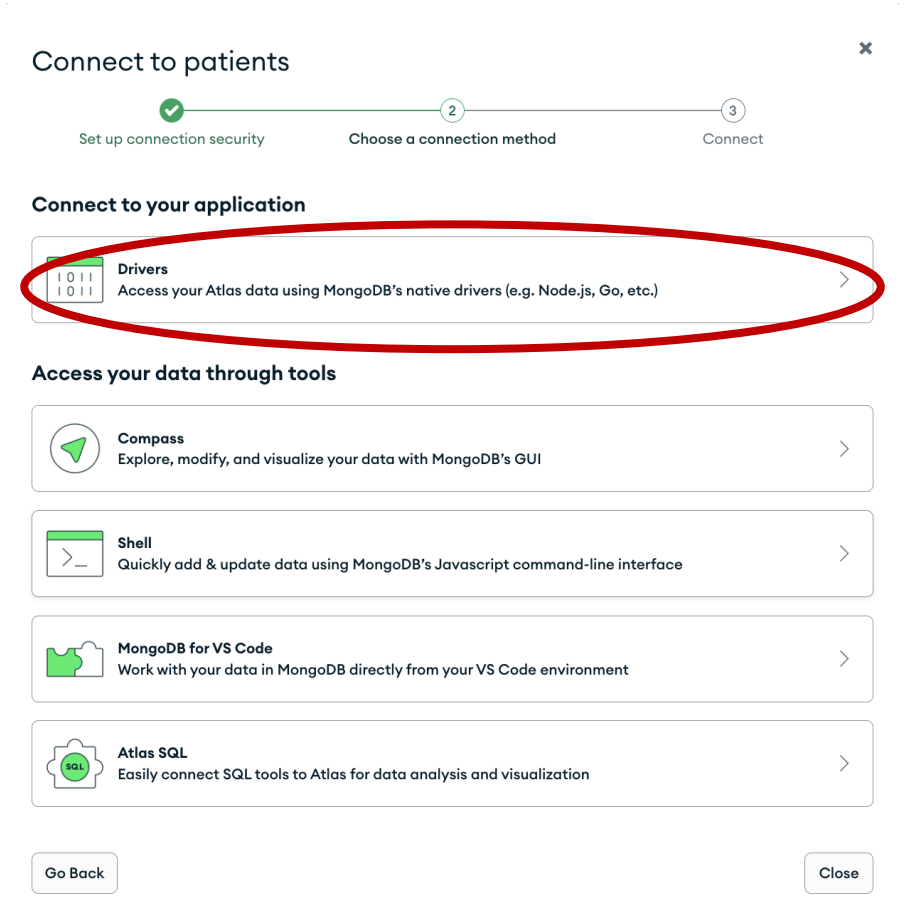

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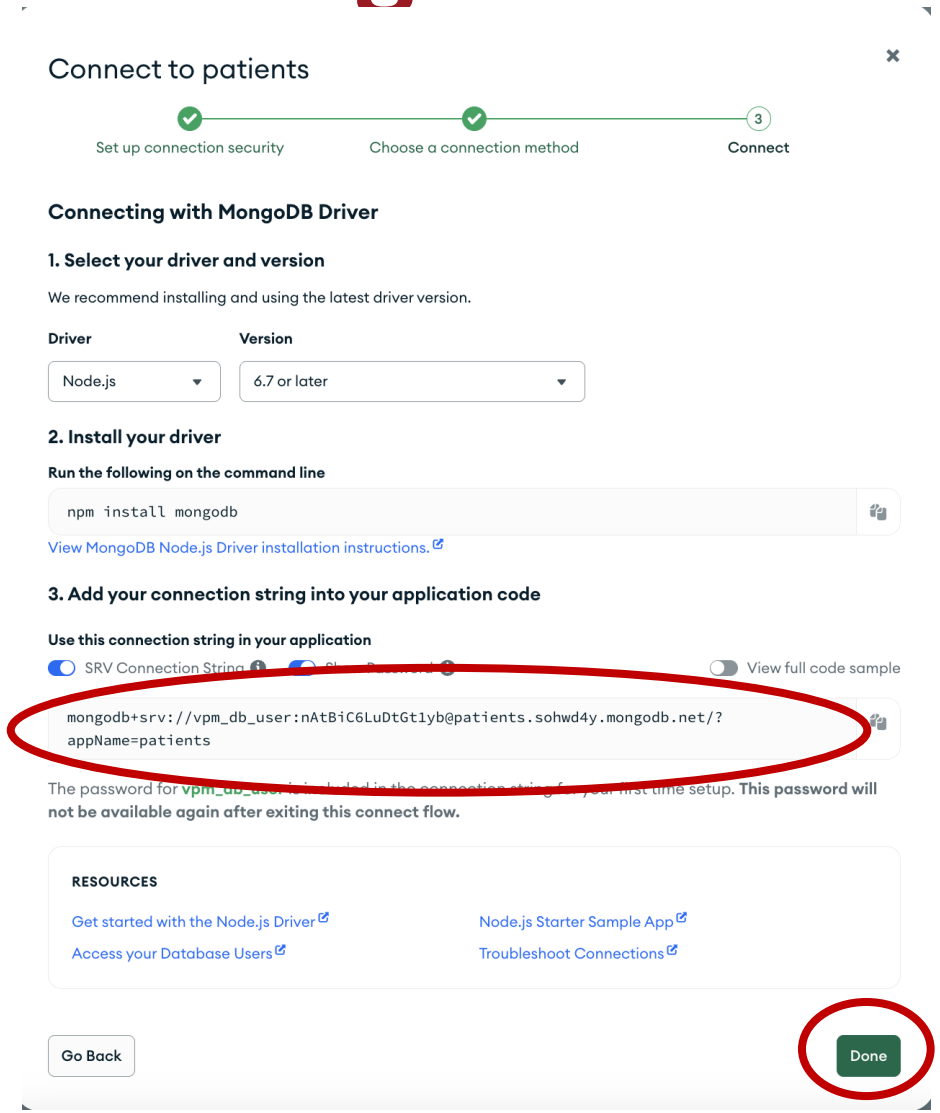
Database – Configure the DB connection



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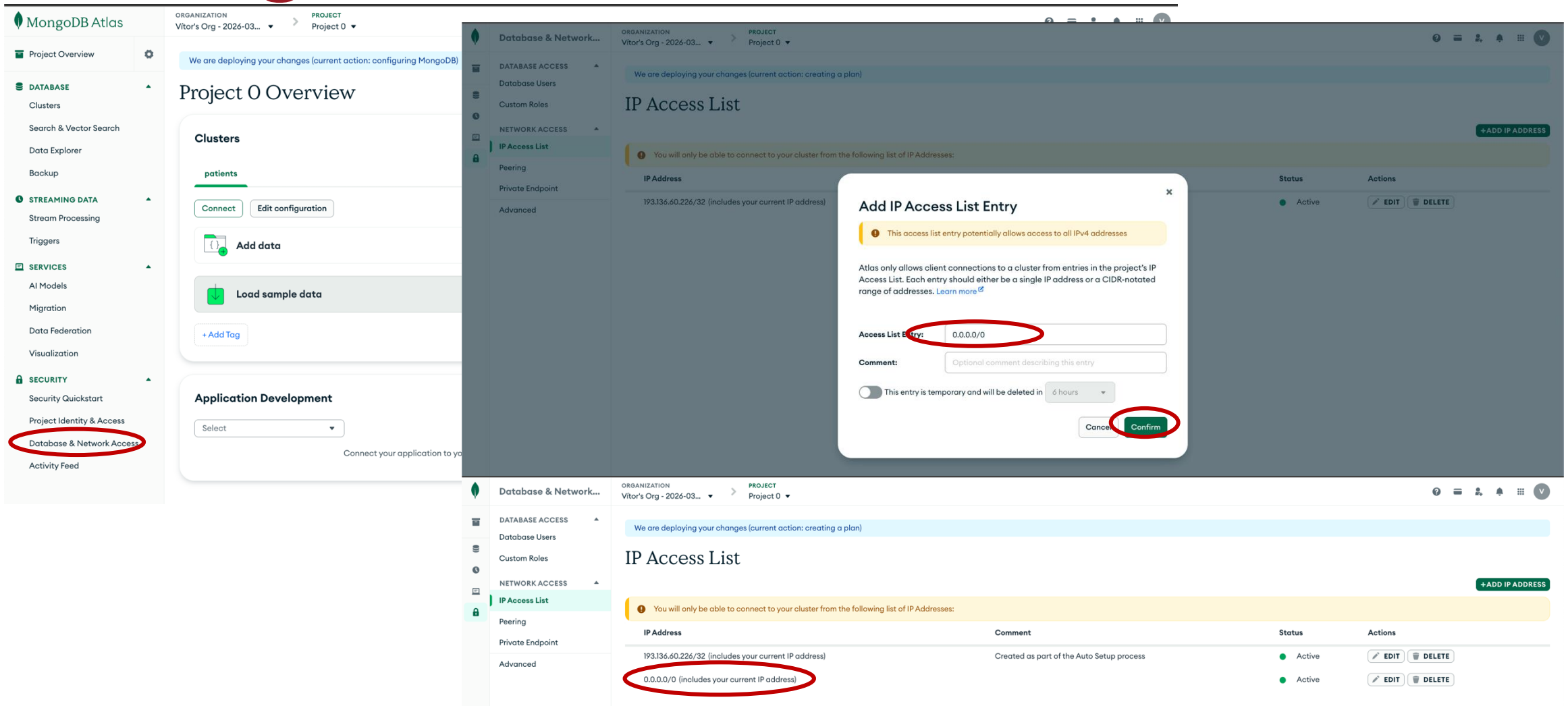
Config IP whitelist



Database

- In the server.js file, replace the example database connection string from the tutorial with your own database connection string.

```
// set our port
var port = process.env.PORT || 8080;
// database
var mongoose   = require('mongoose');
mongoose.connect('mongodb+srv://user123:teste123@patients.xpftma7.mongodb.net/?appName=patients'); // connect to our database
```

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Source Code

- Go to the tutorial and copy the code related to the files server.js and patients.js.
- In the tutorial, the code is presented in stages. Pay attention to the comments to paste the code in the correct place and avoid duplications.
- Or… access the files available on Moodle (folder “Aula_Pratica_4_Codigo”).

Code execution

- To run the code, use the Visual Studio Code terminal.

- If you want to stop it, press “Ctrl + C”.

```
vitorcrista@MacBook-Pro-de-Vitor pl4 % nodemon server.js
[nodemon] 3.1.14
[nodemon] to restart at any time, enter `rs`
[nodemon] watching path(s): *.*
[nodemon] watching extensions: js,mjs,cjs,json
[nodemon] starting `node server.js`
Magic happens on port 8080
Something is happening.
```

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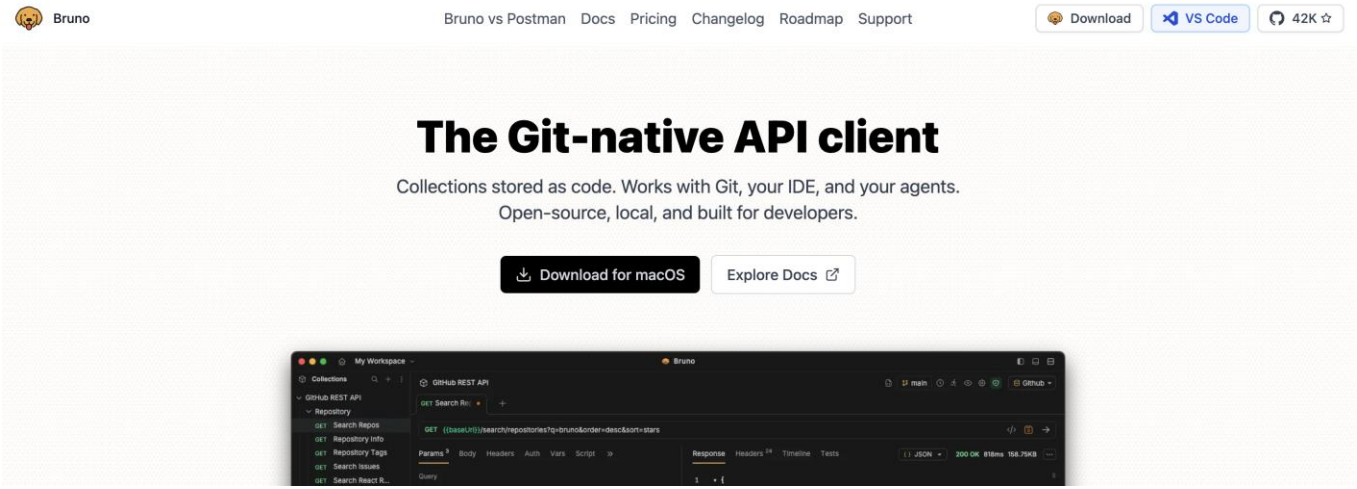
WE ~~DON'T~~ TALK ABOUT BRUNO

AN API CLIENT ALTERNATIVE

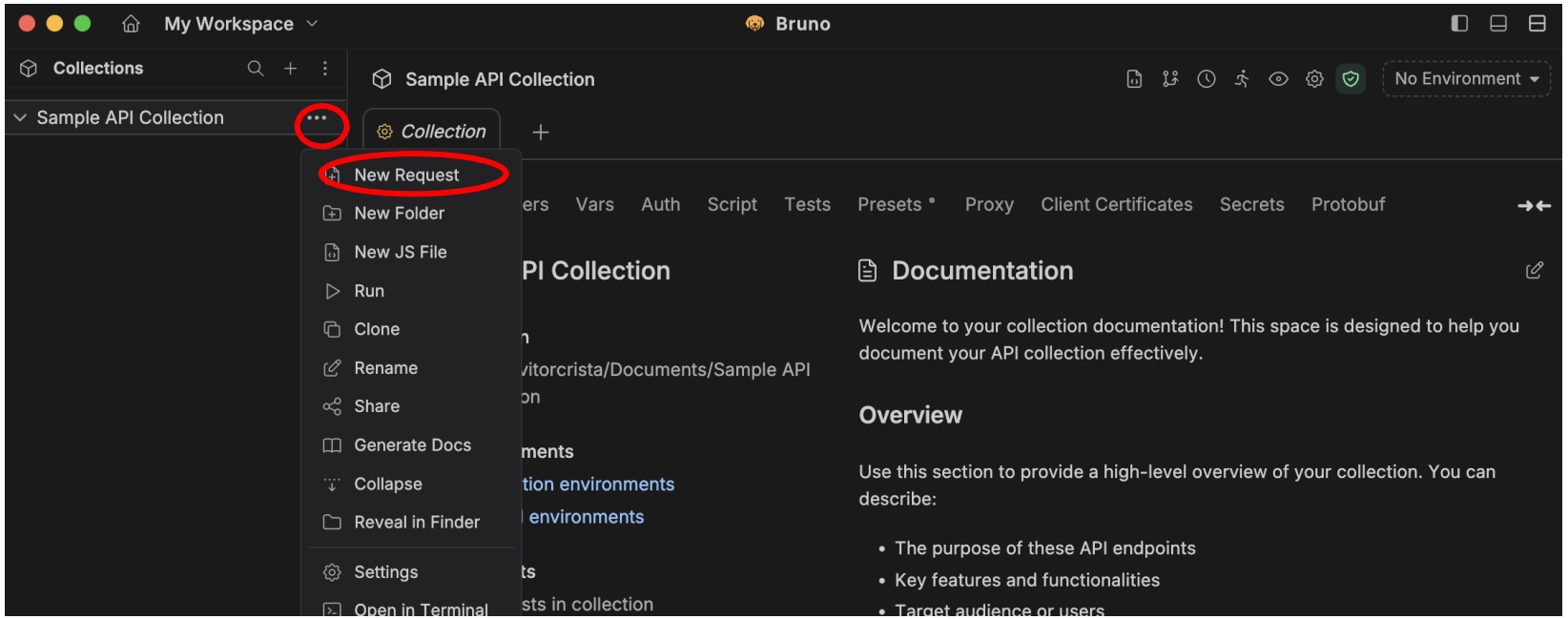
Test through Bruno

- Go to Bruno's website (https://www.usebruno.com/).

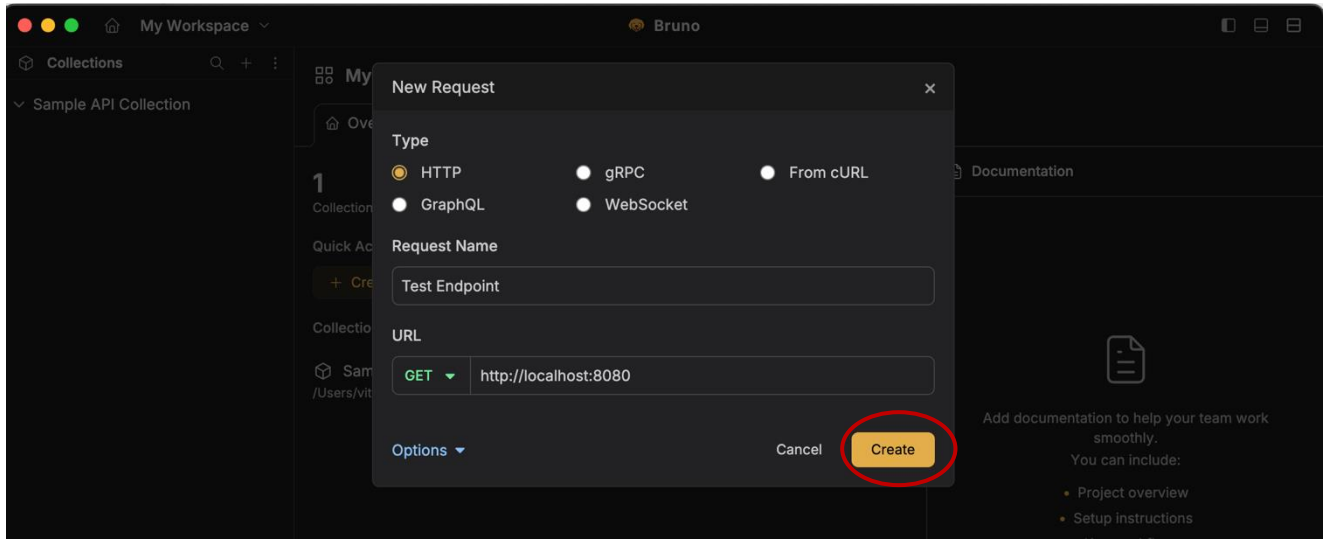


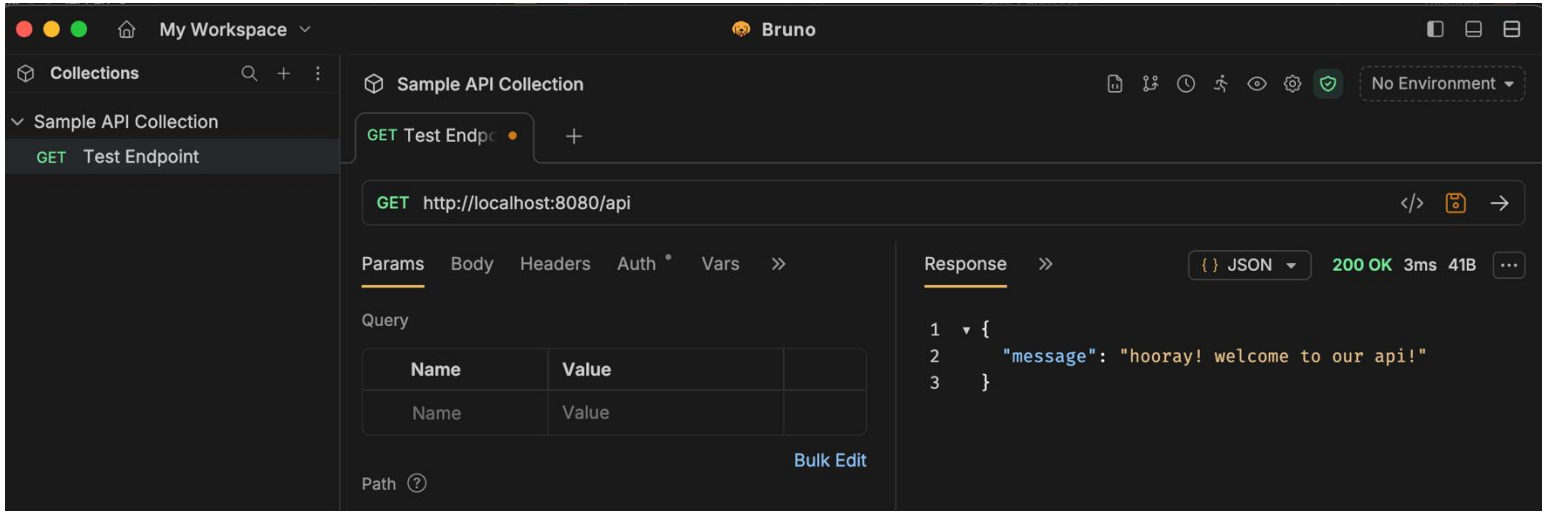
Test through Bruno

- Create a new request



Test through Bruno (cont.)





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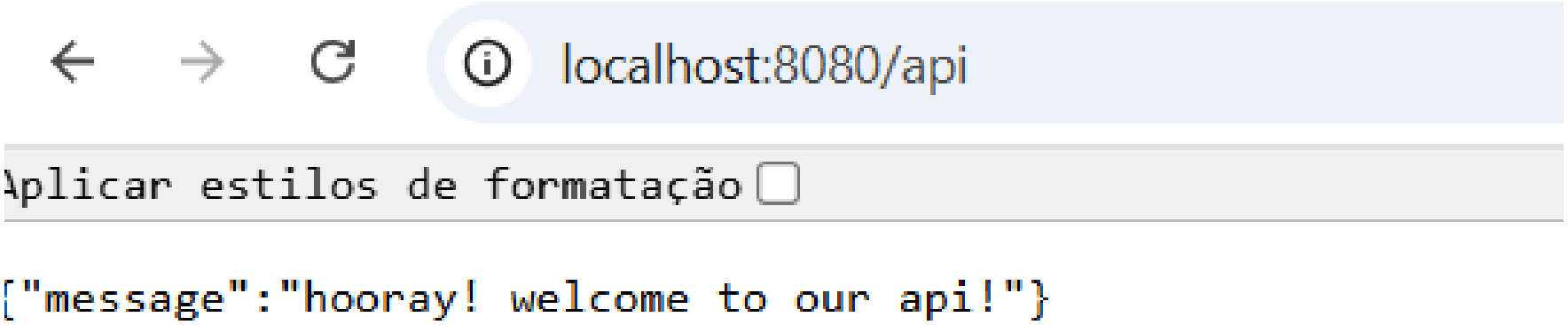
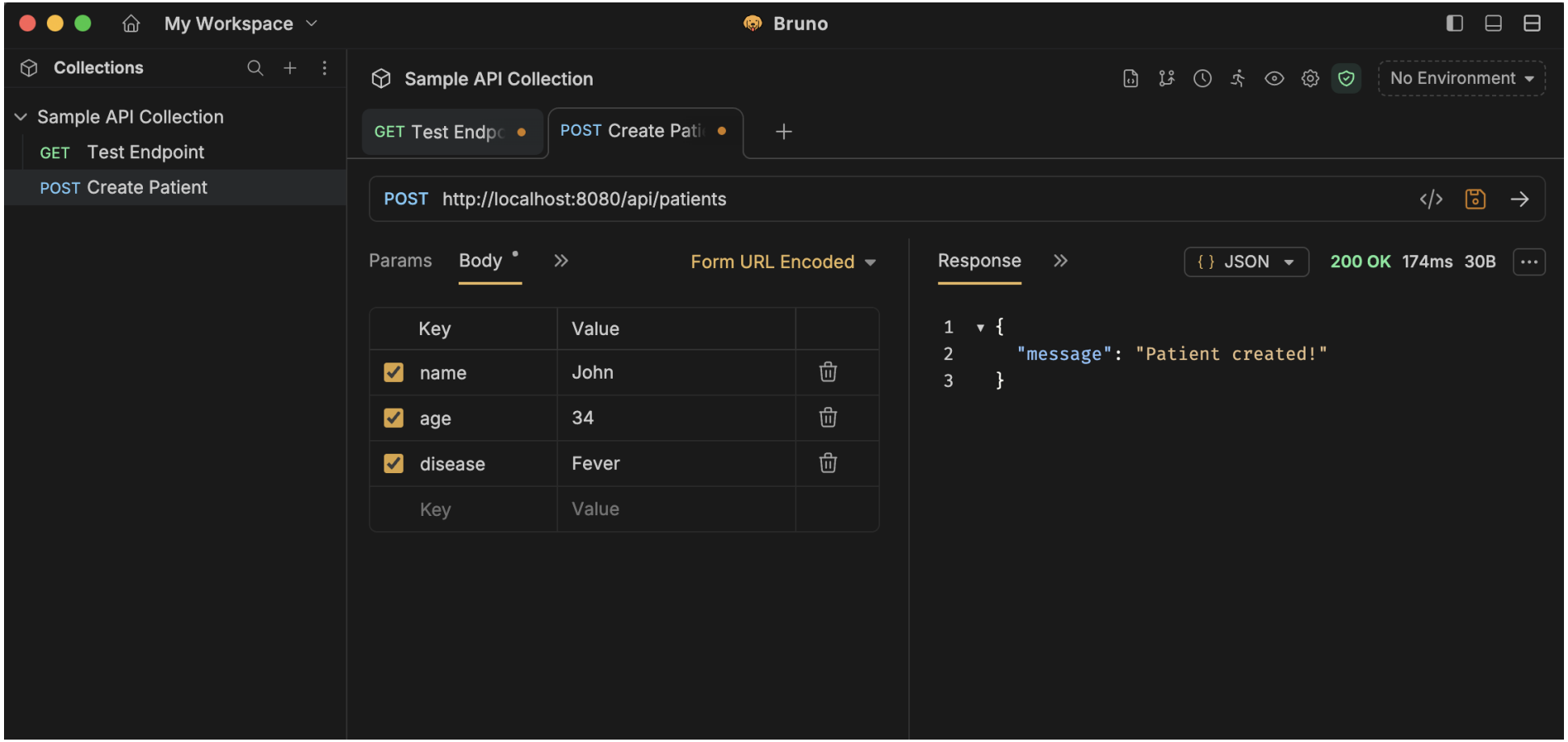
Execution - Browser

- Run `node server.js`



Test with Bruno



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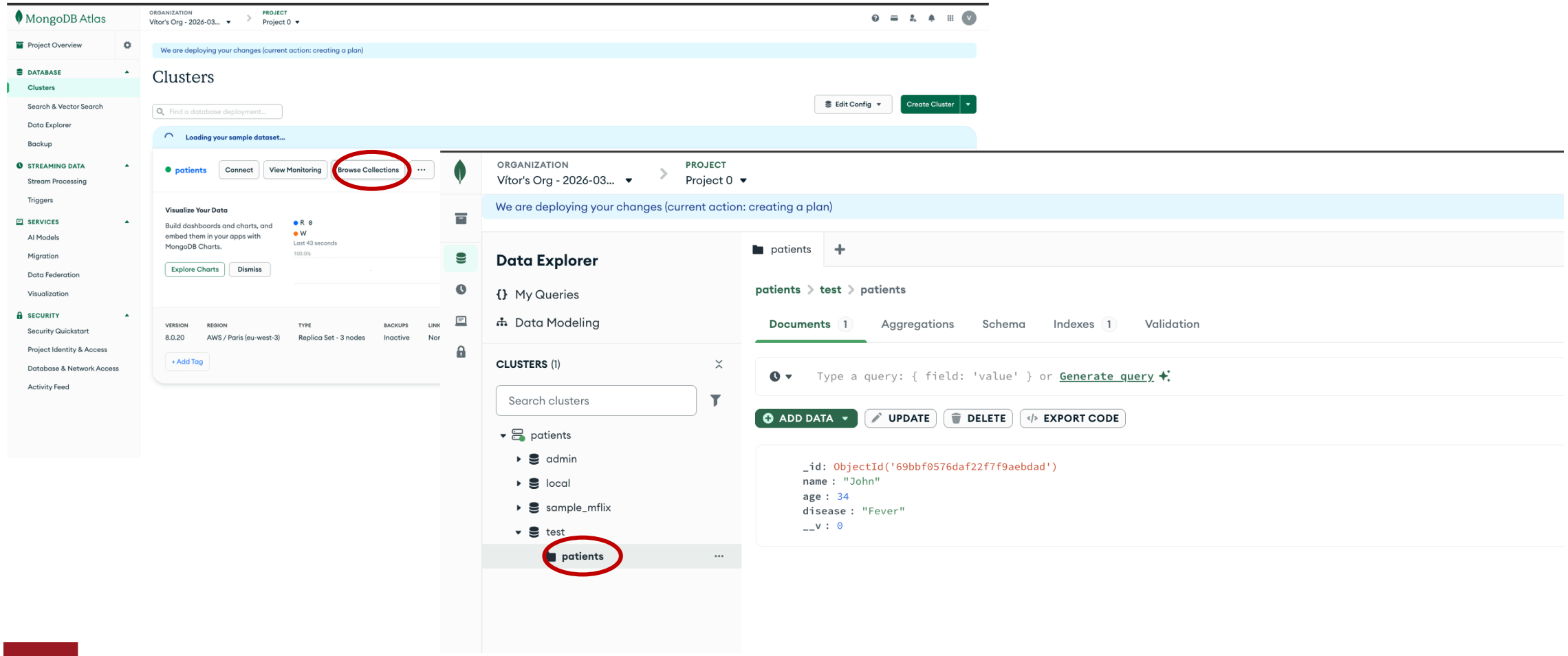
See changes in the Mongodb cloud



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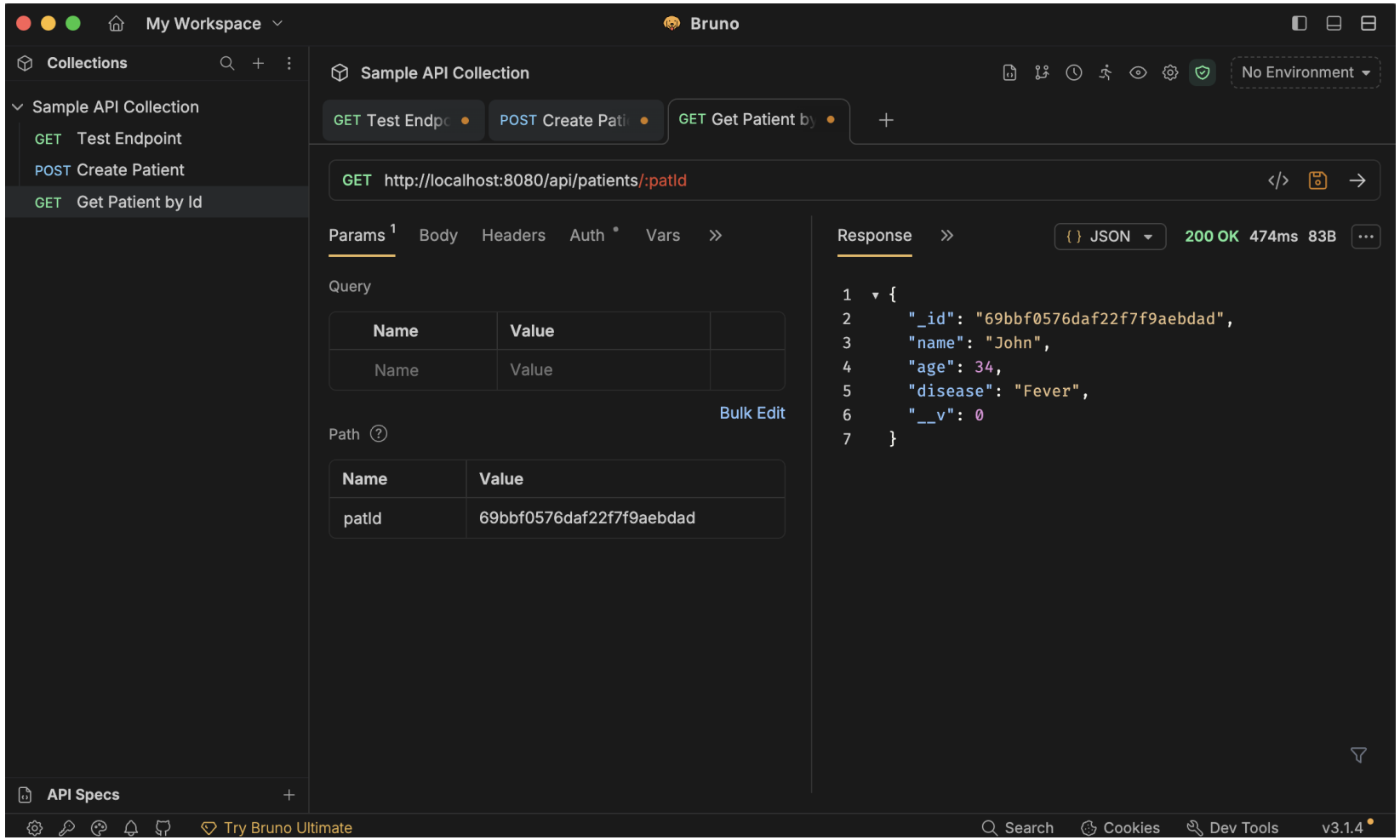
Get patient by id



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Thank You